DS+X 2026 · ART HACK · BOSTON UNIVERSITY

OPEN CANVAS

Explore a painting by touch.

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THE PROBLEM

Art is made to be seen.

2.2 billion

people live with a near or distance vision impairment.

Source: World Health Organization, World report on vision

Paintings can’t be touched.

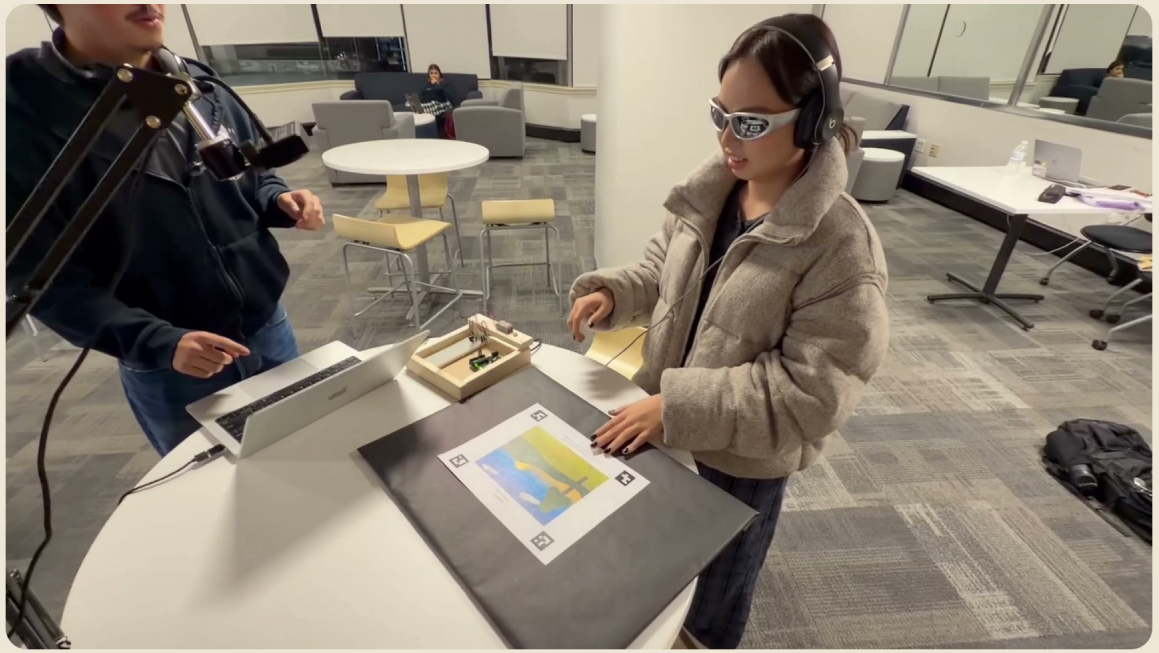
Museums keep hands off the art, so a painting’s detail stays out of reach.

Audio guides are one-way.

One voice, one order, one pace. You hear about a painting; you don’t explore it.

OUR ANSWER

Put a hand on the painting. Hear what's under your fingertip.



Open Canvas turns a printed painting into something blind and low-vision people can explore on their own: in any order, at their own pace.

No screen, menu or keyboard for the visitor. Swap the printed page and it recognizes the new painting.

Demo: a blindfolded tester explores Sunset Hills.

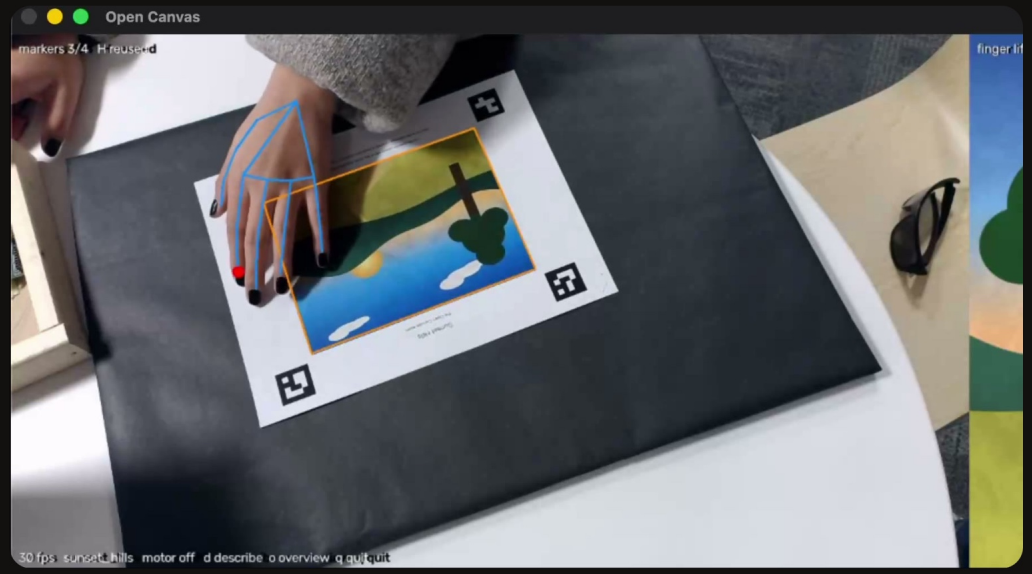
FEATURES

Everything happens under one fingertip

Overview first	A voice names the painting and maps out the scene.
Color becomes music	Blue harp, warm strings, green woodwind, pale choir, dark cello. Brighter plays higher.
Objects make sounds	Leaves rustle in the tree, crickets chirp in the field.
Edges are felt	A fingertip motor buzzes when you cross an outline.
Rest to hear more	Hold still for 2.5 seconds and a voice describes that spot.
Swap the page	Four paintings, each recognized by its printed markers.

HOW IT WORKS

From fingertip to sound, every frame



Live view: a hand hides one marker, so the last good mapping is reused.

01 An overhead webcam films the page at 30 fps.

02 Four ArUco markers identify the painting and map camera pixels onto it.

03 MediaPipe finds the index fingertip.

04 We look up color, region and edge in the clean digital painting.

05 Sound goes to headphones; the edge buzz goes to an Arduino and motor.

TECH STACK

Built from off-the-shelf parts

Vision	OpenCV 5 ArUco markers + homography · MediaPipe hand model
Audio	numpy mixer · sounddevice · ElevenLabs voices and effects
Hardware	Acer webcam · Arduino Uno · transistor driver · coin motor
Launch	Mac app + “Start Open Canvas” Siri Shortcut, no typing
Languages	Python 3.11 with pinned packages · Arduino C++

Kiro: not used

We built with Claude Code instead, working from a shared spec file (CLAUDE.md) that holds the team’s interface contracts.

DATA

Our data and where it came from

Four paintings

Hokusai, Van Gogh, Mondrian (public domain, Wikimedia Commons) + our own.

Region maps we drew

Hand-traced into 6–8 regions each, with a written description and sound.

Live webcam frames

1280×720 at 30 fps. Only the fingertip position is used; nothing is saved.

Pretrained hand model

Google MediaPipe Hand Landmarker (21 keypoints). No training data collected.

How we identified it: only images we may print (public domain or our own); the objects each scene is made of, traced by hand; and design choices grounded in accessibility research (Eyes-Free Art, Rector et al. 2017; EyeMusic) and WHO vision data.

DATA SCIENCE · METHOD 1

Segmentation and soft classification



1 · The painting



2 · Labeled regions



3 · Edge band for the motor

Soft color classes

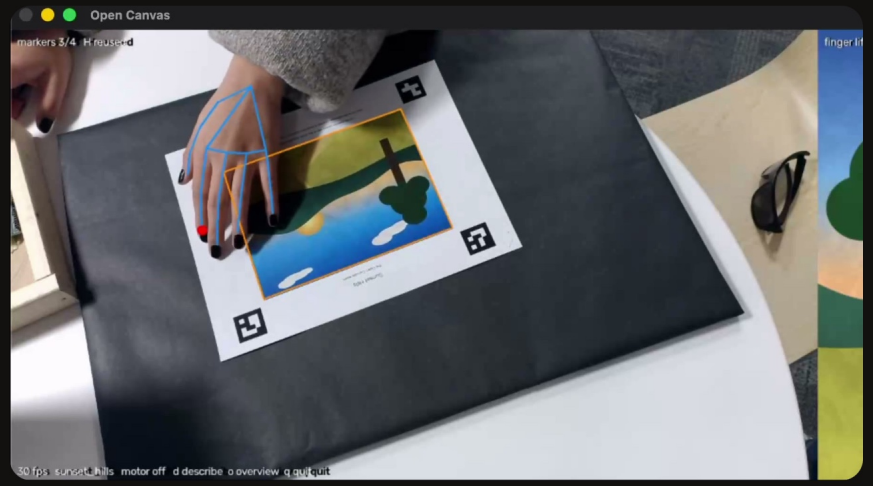
Each color gets weights for five instruments that sum to 1, so sound crossfades at borders instead of flickering.

A data-quality catch

Luma reads navy as 0.06 (black, so cello). HSL lightness reads 0.25 (blue, so harp). We use lightness.

DATA SCIENCE · METHOD 2

Prediction with anomaly rejection



“markers 3/4 · H reused”: the fallback, live.

Predict. A pretrained model finds 21 hand keypoints; we keep the fingertip if confidence ≥ 0.5.

Reject outliers. A non-convex marker shape is a bad detection; we keep the last good mapping.

Smooth. An exponential moving average (α = 0.5) steadies the fingertip.

Debounce. The motor switches after 2 agreeing frames; a description needs 2.5 s of stillness.

IMPACT

Explore art on your own terms

≈$95

in parts we bought (webcam, desk arm, motors), plus a laptop and an Arduino.

4

paintings today. A new one needs an image, a region map and text, not new code.

0

screens or menus for the visitor. Swap the page; Siri starts the app.

Instead of someone else's tour, visitors choose where to go, in any order, at their own pace.

So far tested with blindfolded sighted testers. Next: blind and low-vision users.

ETHICS & RESPONSIBILITY

Our statement

Open Canvas is for blind and low-vision people, so its mistakes cost them more than us. We print only public-domain paintings or our own, credit ElevenLabs for generated voices and sounds, and use Google's pretrained hand model without collecting training data. The camera looks down at the table; frames stay in memory, are never saved or uploaded, and only the fingertip position is used. Hand tracking may work less well for some skin tones, lighting or hand differences, which we have not yet measured; our descriptions carry a sighted team's reading of mostly Western art; and the system assumes hearing, English and a usable fingertip. Because a wrong description misleads someone who can't check it, every spoken line is written and reviewed by us, not generated live. The overview explains the system before exploring, the README lists sources and limits, and so far only blindfolded sighted people have tested it. Blind and low-vision users must shape what comes next.

Sourcing, privacy and fairness

Responsible sourcing & use

- Paintings are public domain (Wikimedia Commons) or our own.
- We drew every region map and wrote every description.
- ElevenLabs voices and effects, credited as its free plan requires.
- Research and the WHO figure are cited by name.

Privacy, bias & fairness

- The camera points at the table; frames are never stored or sent.
- Hand tracking may vary with skin tone, lighting or limb differences; not yet measured.
- Descriptions reflect a sighted team’s view of mostly Western art.
- Excludes deafblind users, non-English speakers, and anyone who can’t use a fingertip.

ETHICS · HARMS & MITIGATION

What could go wrong, and what we did

Possible harm	What we did
A wrong description misleads someone who can’t check it	Every line is written and reviewed by us; no live AI answers in the demo
Sensory overload or being startled	Quiet music bed that ducks under speech; descriptions only after a 2.5 s rest
Motor chatter on the fingertip	Hysteresis; a low-voltage coin motor behind a transistor and flyback diode
Silent failure when tracking is lost	Last good mapping is reused; when the finger is lost, the motor turns off

ETHICS · TRANSPARENCY & ACCOUNTABILITY

Clear about what it is and isn’t

01 The spoken overview names the painting and explains how to explore before anything else plays.

02 Every description is pre-written by the team, so what users hear can be checked line by line.

03 The README and CREDITS list data sources, licenses, methods and limits; all code is public on GitHub.

04 Limits stated up front: not yet tested with blind or low-vision users, tracking accuracy across hands not measured, four paintings only.

NEXT STEPS

What’s next, and thank you

- Test with blind and low-vision users; let their feedback set priorities.
- Measure tracking accuracy across skin tones, lighting and hands.
- Optional live descriptions for new paintings, with human review.
- A wearable fingertip mount, more paintings, more languages.

LINKS

github.com/bee-0207/open-canvas

Watch the 1:42 video

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Voices and sound effects by ElevenLabs · Paintings: Wikimedia Commons (public domain) · Research: Eyes-Free Art (Rector et al., Microsoft Research, 2017), EyeMusic